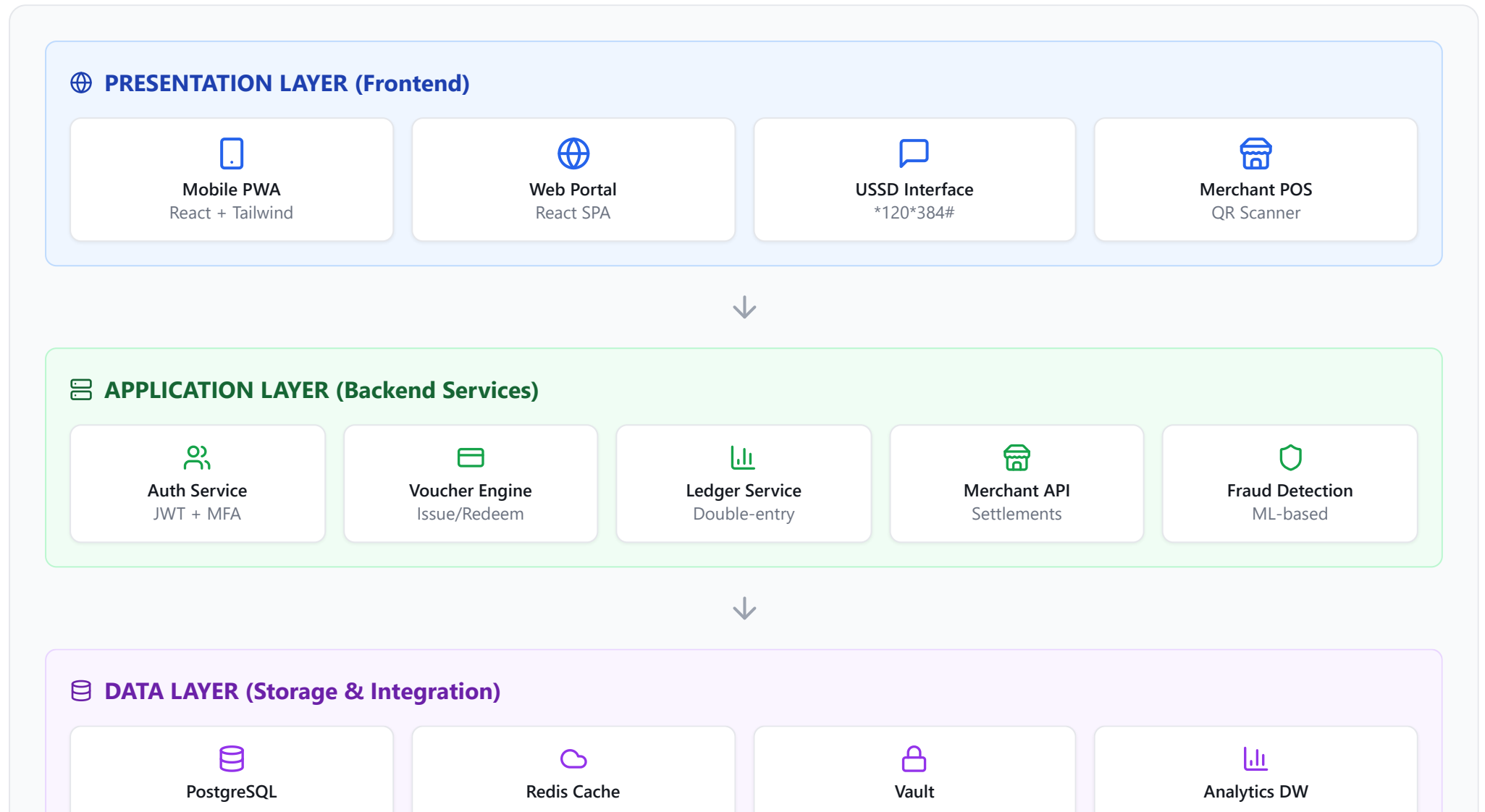




1. High-Level System Architecture



Primary DB

Session/Cache

Secrets Mgmt

Reporting

🔗 2. Technology Stack

Frontend

React 18
UI Framework

Tailwind CSS
Styling

React Query
Data Fetching

Framer Motion
Animations

PWA
Offline Support

Backend

Node.js / Python
API Services

RESTful API
Communication

JWT + OAuth2
Authentication

Serverless Functions
Microservices

Queue System
Async Processing

Infrastructure

AWS / Azure
Cloud Platform

PostgreSQL
Database

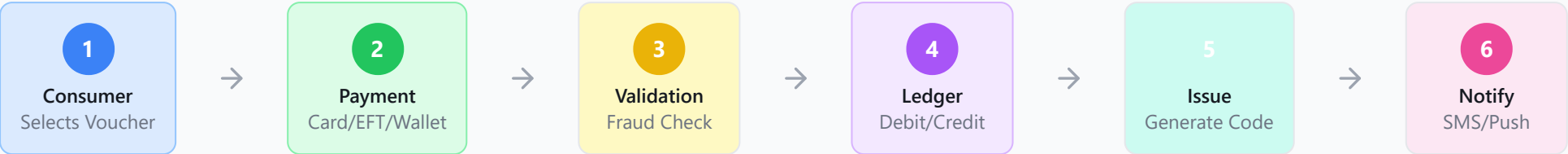
Redis
Caching Layer

CDN
Content Delivery

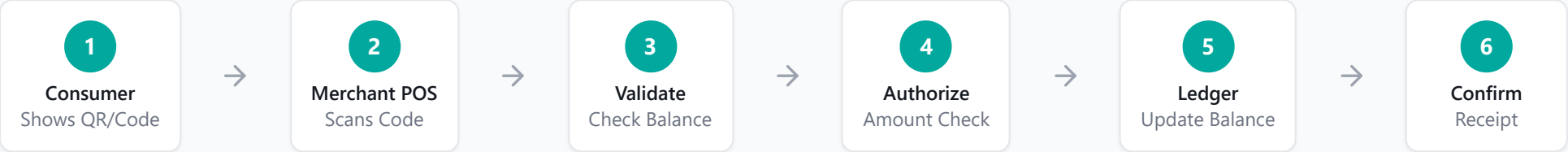
Docker + K8s
Containerization

🔗 3. Transaction Data Flow

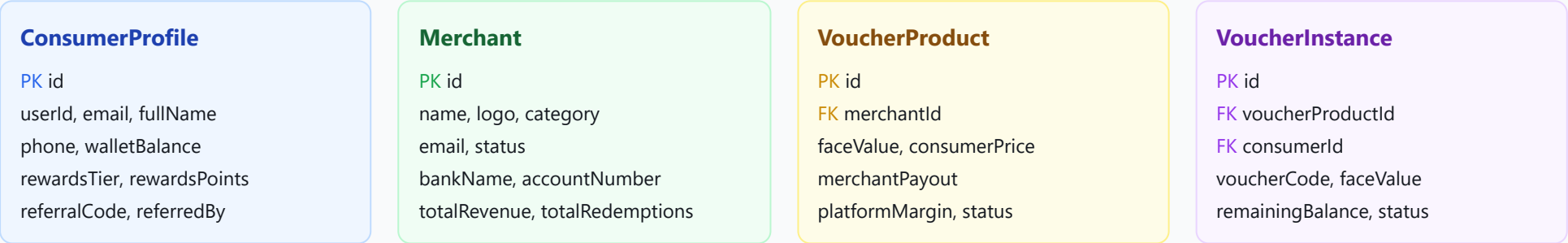
3.1 Voucher Purchase Flow



3.2 Voucher Redemption Flow



🗄️ 4. Entity Relationship Diagram



Transaction

PK id
type, amount, status
FK userId, merchantId
paymentMethod, reference
description

LedgerEntry

PK id
entryType, amount
FK transactionId
FK merchantId
reference, description

WalletTransaction

PK id
FK userId
type, amount
balanceAfter, reference
recipientPhone, status

Referral

PK id
FK referrerUserId
FK referredUserId
referralCode, bonusAmount
status, completedDate

Relationships:

- Merchant (1) → (*) VoucherProduct
- ConsumerProfile (1) → (*) VoucherInstance
- Transaction (1) → (*) LedgerEntry
- VoucherProduct (1) → (*) VoucherInstance
- ConsumerProfile (1) → (*) Transaction
- ConsumerProfile (1) → (*) WalletTransaction

5. Security Architecture

Security Layers

- 1 Network Security**
TLS 1.3, WAF, DDoS Protection, VPN
- 2 Application Security**
JWT Auth, RBAC, Input Validation, OWASP
- 3 Data Security**
AES-256 Encryption, Key Rotation, Masking
- 4 Operational Security**
Audit Logs, SIEM, Incident Response

Compliance Framework

- | | |
|---------------------------------|-----------------------------------|
| POPIA
Data Protection | PASA
Payment Standards |
| SARB
Reserve Bank | FIC
Financial Intel |
| PCI-DSS
Card Security | ISO 27001
Info Security |

6. External Integration Points

Payment Gateways

-  Peach Payments
-  PayFast
-  Ozow (EFT)
-  SnapScan

Communication

-  USSD Gateway (*120*384#)
-  SMS API (Clickatell)
-  Push Notifications
-  Email (SendGrid)

Third Party

-  Retailer POS Systems
-  SASSA Database
-  Analytics (Google)
-  KYC Provider



eVoucher Platform © 2025
3P's Social Business Model

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